Trigonometric integrals (7.2)

This section covers methods to evaluate integrals of the forms

$$\int \sin^m x \cos^n x \, dx \qquad \int \tan^m x \sec^n x \, dx$$

where m, n are nonnegative integers.

The main tools we will use are trigonometric identities and the substitution rule.

Useful trig identities:

$\sin^2 x + \cos^2 x = 1$ (Pythagorean theorem)

$\tan^2 x + 1 = \sec^2 x$

$\sin 2x = 2\sin x \cos x$

$\cos 2x = 1 - 2\sin^2 x$

$\cos 2x = 2\cos^2 x - 1$

$\sin x \cos x = \frac{1}{2}\sin 2x$

$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

$\int \sin^m x \cos^n x \, dx$ where **m is odd** (n anything)

Use $\sin^2 x = 1 - \cos^2 x$ to rewrite $\sin^m x$ in terms of $\cos x$, keeping one factor of $\sin x$.

Substitute $u = \cos x$.

$$\int \sin^3 x \cos^2 x \, dx \qquad \sin^2 x = 1 - \cos^2 x$$

$$\int \sin x \sin^2 x \cos^2 x \, dx$$

$$u = \cos x$$

$$\int \sin x \, (1 - \cos^2 x) \cos^2 x \, dx \qquad du = -\sin x \, dx$$

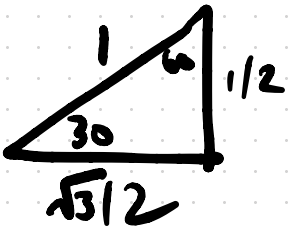
$$\int (1 - u^2) u^2 \, (-du) = \int (u^2 - u^4)(-du)$$

$$\int (u^4 - u^2) \, du = \frac{1}{5} u^5 - \frac{1}{3} u^3 + C$$

$$= \frac{1}{5} \cos^5 x - \frac{1}{3} \cos^3 x + C$$

$$\int_0^{\pi/6} \sin^5 x \, dx$$

$$\sin^2 x = 1 - \cos^2 x$$



$$\int_0^{\pi/6} \sin x \, (\sin^2 x)^2 \, dx$$

$$\int_0^{\pi/6} \sin x \, (1-\cos^2 x)^2 \, dx$$

$$u = \cos x$$

$$du = -\sin x \, dx$$

$$\int_1^{\sqrt{3}/2} (1-u^2)^2 (-du)$$

$$\int_{\sqrt{3}/2}^{1} (1-u^2)^2 \, du = \int_{\sqrt{3}/2}^{1} (1 - 2u^2 + u^4) \, du$$

$$= u - \frac{2}{3}u^3 + \frac{1}{5}u^5 \Big|_{\sqrt{3}/2}^{1}$$

$$= \left(1 - \frac{2}{3} + \frac{1}{5}\right) - \left(\frac{\sqrt{3}}{2} - \frac{2}{3}\cdot\frac{3\sqrt{3}}{8} + \frac{1}{5}\cdot\frac{9\sqrt{3}}{32}\right)$$

$\int \sin^m x \cos^n x \, dx$ where n is odd (m anything)

Use $\cos^2 x = 1 - \sin^2 x$ to rewrite $\cos^n x$ in terms of $\sin x$, keeping one factor of $\cos x$.

Substitute $u = \sin x$.

(If both m and n are odd, you can use either method.)

$$\int_0^{\pi/6} \cos^3 x \, dx \qquad \cos^2 x = 1 - \sin^2 x$$

$$\int_0^{\pi/6} \cos x \, \cos^2 x \, dx$$

$$\int_0^{\pi/6} \cos x \, (1 - \sin^2 x) \, dx \qquad u = \sin x \qquad du = \cos x \, dx$$

$$\int_0^{1/2} (1 - u^2) \, du$$

$$u - \frac{1}{3}u^3 \Big|_0^{1/2} = \frac{1}{2} - \frac{1}{24} = \frac{11}{24}$$

$$\int \sin^2(\pi x)\cos^3(\pi x)\,dx \qquad \cos^2(\pi x) = 1 - \sin^2(\pi x)$$

$$\int \sin^2(\pi x)\cos(\pi x)\cos^2(\pi x)\,dx$$

$$\int \sin^2(\pi x)\cos(\pi x)(1-\sin^2(\pi x))\,dx$$

$$u = \sin(\pi x)$$

$$du = \pi\cos(\pi x)\,dx$$

$$\frac{1}{\pi}du = \cos(\pi x)\,dx$$

$$\int u^2(1-u^2)\cdot\left(\frac{1}{\pi}du\right)$$

$$\frac{1}{\pi}\int (u^2-u^4)\,du$$

$$\frac{1}{\pi}\left(\frac{1}{3}u^3-\frac{1}{5}u^5\right)+C = \frac{1}{\pi}\left(\frac{1}{3}\sin^3(\pi x)-\frac{1}{5}\sin^5(\pi x)\right)+C$$

$\int \sin^m x \cos^n x \, dx$ where m, n are both even

Use the identities

$$\sin x \cos x = \frac{1}{2} \sin 2x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

to reduce powers.

$$\int \cos^2 x \, dx \qquad \cos^2 x = \frac{1}{2}(1+\cos 2x)$$

$$\int \frac{1}{2}(1+\cos 2x)\, dx$$

$$\frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) + C$$

$$\int \sin^2 x \cos^2 x \, dx$$

Method 1:

$$\int \sin^2 x \,(1-\sin^2 x)\, dx \qquad \sin^2 x = \frac{1}{2}(1-\cos 2x)$$

$$\int \frac{1}{2}(1-\cos 2x)\left(1-\frac{1}{2}(1-\cos 2x)\right) dx$$

$$\int \frac{1}{2}(1-\cos 2x)\left(\frac{1}{2}+\frac{1}{2}\cos 2x\right) dx$$

$$\int \frac{1}{4}(1-\cos 2x)(1+\cos 2x)\, dx$$

$$\int \frac{1}{4}(1-\cos^2 2x)\,dx \qquad \cos^2 2x = \frac{1}{2}(1+\cos 4x)$$

$$\int \frac{1}{4}\left(1-\frac{1}{2}(1+\cos 4x)\right)dx$$

$$\int \frac{1}{4}\left(\frac{1}{2}-\frac{1}{2}\cos 4x\right)dx$$

$$\int \frac{1}{8}(1-\cos 4x)\,dx$$

$$\frac{1}{8}\left(x-\frac{1}{4}\sin 4x\right)+C$$

$$\int \sin^2 x \cos^2 x \, dx$$

Method 2:

$$\sin x \cos x = \frac{1}{2} \sin 2x$$

$$\int (\sin x \cos x)^2 \, dx$$

$$\int \frac{1}{4} \sin^2 2x \, dx$$

$$\sin^2 2x = \frac{1}{2}(1 - \cos 4x)$$

$$\int \frac{1}{8}(1 - \cos 4x) \, dx$$

$$\frac{1}{8}\left(x - \frac{1}{4} \sin 4x\right) + C$$

$$\int \tan^m x \sec^n x \, dx = \int \frac{\sin^m x}{\cos^{m+n} x} \, dx$$

$\frac{d}{dx} \tan x = \sec^2 x$ $\quad$ $\frac{d}{dx} \sec x = \sec x \tan x$

$\tan^2 x + 1 = \sec^2 x$

Try to do one of the following

(a) Convert $\sec^2 x$ to $\tan^2 x + 1$ except one factor of $\sec^2 x$, substitute $u = \tan x$.

(b) Convert $\tan^2 x$ to $\sec^2 x - 1$ except one factor of $\sec x \tan x$, substitute $u = \sec x$.

$$\int_0^{\pi/4} \tan^3 x \sec^4 x \, dx$$

Method (a): $\sec^2 x = \tan^2 x + 1$

$$\int_0^{\pi/4} \tan^3 x \sec^2 x \sec^2 x \, dx$$

$$\int_0^{\pi/4} \tan^3 x \sec^2 x (\tan^2 x + 1) \, dx$$

$u = \tan x$

$du = \sec^2 x \, dx$

$$\int_0^1 u^3 (u^2 + 1) \, du$$

$$\int_0^1 (u^5 + u^3) \, du = \frac{1}{6} u^6 + \frac{1}{4} u^4 \Big|_0^1 = \frac{1}{6} + \frac{1}{4} = \frac{5}{12}$$

$$\int \tan x \, dx$$

$$\int \frac{\sin x}{\cos x} \, dx$$

$$u = \cos x$$

$$du = -\sin x \, dx$$

$$\int -\frac{1}{u} \, du$$

$$-\ln|u| + C$$

$$-\ln|\cos x| + C$$

$$\ln|\sec x| + C$$

$$\int \sec x \, dx$$

$$\int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} \, dx$$

$$\int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx$$

$$u = \sec x + \tan x$$

$$du = (\sec^2 x + \sec x \tan x) \, dx$$

$$\int \frac{1}{u} \, du$$

$$\ln|u| + C = \ln|\sec x + \tan x| + C$$